

Biogenic carbon capture at pulp mills via sodium spiking and oxy-fuel calcination

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ABSTRACT

Over 13 million metric tons of biogenic carbon dioxide (CO₂) are mineralized yearly in United States (US) pulp mill recovery boilers as molten sodium carbonate. The mineralized CO₂ is released downstream in the rotary lime kiln where it can be captured at a relatively low cost for permanent sequestration. The use of biocarbon and bioenergy at pulp mills enables the possibility for atmospheric carbon removal when biogenic CO₂ is captured and sequestered. We demonstrate the feasibility of capturing biogenic CO₂ via sodium spiking coupled with oxy-fuel calcination in the rotary lime kiln at existing kraft pulp mills. Sodium (Na) spiking elevates the Na ion loading in the kraft chemical looping process by replacing chlorine-based bleaching with a highly alkaline bleaching sequence. Chemical pulping processes are modeled and simulated to understand the technical limits of implementing sodium spiking in existing pulp mills. Each 1 % of sodium added to the bleaching operations increases the rate of CO₂ mineralization by 4 %, and the maximum increase in sodium content in the kraft process is 9 %. Without sodium spiking, estimated CO₂ capture costs are \$131 and \$107/mt CO₂ for air and oxy-fuel combustion, respectively. With the implementation of sodium spiking, the cost of CO₂ captured decreases by 27 % and 31 %, showing costs of \$96 and \$74/mt CO₂ for air and oxy-fuel combustion, respectively. Oxy-fuel combustion reduces the costs of CO₂ capture compared to air combustion, but merged deployment with sodium spiking seems to be the most cost-effective pathway to integrate carbon capture.

1. Introduction

Many countries have committed to achieving net zero greenhouse gas emissions by 2050, requiring significant deployment of carbon dioxide removal technologies to account for hard-to-abate emissions, such as those from the industrial sector. A variety of important commodities originate from the pulp and paper industry such as packaging, clothing, food, and hygiene products, all of which make everyday life both possible and more convenient. Due to the present and growing scale of the pulp and paper industry, this sector of the industrial economy will play a significant role in CO₂ removal due to the unique opportunity for large-scale biogenic CO₂ capture and sequestration. In the US, approximately 110 million tonnes of biogenic CO₂ are emitted annually from existing pulp mills (Sagues et al., 2020). Integrating CO₂ capture technologies into existing biomass pulping processes has the potential for near-term impact. However, challenges must be overcome, including the need to avoid disrupting current operations. In addition, there are multiple sources of biogenic CO₂ at pulp mills, and they must be

prioritized based on economic viability. The primary sources of biogenic CO₂ include recovery boilers (RB), biomass boilers, multi-fuel boilers, and lime kilns (LK). While the recovery boiler represents the highest quantity of CO₂ emissions, the capture of CO₂ is costly due to the various gases involved in significant reactions inside the boiler and the general reluctance from the pulping industry to interfere with recovery boiler operations; the recovery boiler is the most expensive and sensitive operation at most biomass pulping mills. The lime kiln represents the lowest quantity of CO₂ emissions but has several advantages for carbon capture, including its high concentration of CO₂, low amount of impurities, ability to be retrofitted to oxy-fuel, and ability to operate at higher temperatures.

Lime kilns are used in biomass pulping mills to recycle chemicals, as shown in Fig. 1, and might be the most effective pathway for incorporating carbon capture technologies into an integrated pulp manufacturing process. The flue gas (weight basis) emitted by the lime kiln primarily consists of CO₂ (20%), water (12%), residual oxygen (3%), and a significant amount of nitrogen (65%) that enters the kiln

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with the air used for fuel combustion. Replacing air with pure oxygen or oxy-fuel combustion has been proposed to eliminate the presence of nitrogen, reducing the costs of downstream CO₂ capture due to the higher CO₂ concentration reached (Sifat and Haseli, 2019). There are few published studies on integrating oxy-fuel combustion into biomass pulp mills (Lefvert and Grönkvist, 2023), unlike other industrial processes such as power generation (Stanger et al., 2015), (Chen et al., 2019), (Liu, Zhao et al., 2023), cement (Williams et al., 2024), (Meijssen et al., 2024), (Bacatelo et al., 2023), or steel (Hammerschmid et al., 2021), (Perpiñán et al., 2021), where literature on oxy-fuel combustion is more prevalent. Simulations of oxy-fuel combustion coupled with rotary lime kilns have shown that it is possible to produce lime with lower specific energy consumption and capture CO₂ emissions with subsequent transport and storage without affecting lime quality (Eriksson et al., 2014). In addition, techno-economic modeling has indicated that the capture of CO₂ from rotary lime kilns has the potential to be economically viable if existing kilns can be retrofitted (Sagues et al., 2020). In addition, it has been shown that retrofitting lime kilns to oxy-fuel increases the capacity, allowing for higher throughput of lime mud (CaCO₃) (Granados et al., 2015), (Wu et al., 2024). Novel systems and approaches that utilize oxy-fuel combustion have been proposed for limestone calcination (Smadi et al., 2023). However, a large-scale demonstration has yet to be done.

It is essential to note that most studies have focused on applying oxy-fuel combustion in lime kilns used in the cement industry. Most of the advancements made in decarbonizing lime kilns in the cement industry can be used and adapted to lime kilns in the pulping industry, however, the nuances of lime processing in the pulping industry require more dedicated studies that assess lime kiln decarbonization strategies specific to pulp mills (Wu et al., 2024). Prior studies have focused on the advantages of oxy-fuel combustion for integrating complementary carbon capture technologies, which has led to a deeper understanding of how oxy-fuel combustion can enhance CO₂ capture efficiency and reduce energy consumption across various industrial processes (Talei et al., 2024). Talei et al. discuss the advantages of coupling oxy-fuel combustion with an amine absorption carbon capture system compared to traditional post-combustion CO₂ capture for five different flue gases from five different industries such as gas and coal-fired power plants, combined heat and power plants, aluminum manufacturing processes and the cement industry. Their simulation findings showed that the oxy-fuel combustion cases require less energy, mainly attributed to the mass transfer driving force of the higher CO₂ concentration achieved in the inlet flue gas (Talei et al., 2024). In addition, Dubois et al.

evaluated the effect of partial oxy-fuel combustion in a cement kiln by applying a post-combustion CO₂ capture approach using chemical absorption with different blends of amines. They observed an experimental relationship between the increase in CO₂ concentration and the reduction in energy utilization for solvent regeneration. For a typical monoethanolamine (MEA) 30 wt% system, that observation was confirmed through simulations, finding that a 24% increase in CO₂ concentration led to a 26% reduction in energy consumption and up to 37% when the concentration of CO₂ was increased up to 60% (Dubois et al., 2017). However, there is still much to explore regarding its application in biomass pulp mills. In addition, future studies should include the energy demand for oxygen (O₂) production to provide a comprehensive energy use analysis in oxy-fuel systems integration. Notably, many biomass pulp mills already separate high-purity oxygen from air for use in bleaching, and thus existing infrastructure could potentially be leveraged in oxy-fuel applications.

Besides oxy-fuel combustion, electrification is another option for generating high-purity CO₂ for subsequent sequestration. Electrification has gained significant attention as a way to decarbonize the industrial sector (Lechtenböhmer et al., 2016), (Nagovnak et al., 2024). Furthermore, significant research and development have been done on both oxy-fuel and electrified kilns for atmospheric CO₂ capture (DAC) (Keith et al., 2018). Electrified kilns are being tested at a pilot scale to increase CO₂ purity and lower the cost of integrating complementary carbon capture technologies such as DAC. At the same time, cement producers are making efforts to retrofit current kilns to electrified kilns (Liu et al., 2023). In 2021, 20% of the generated electricity in the US came from renewable sources (e.g. wind and solar energy) and is expected to continue increasing, and thus electrification might be a viable way to decarbonize lime kilns in the future (Motalebi et al., 2022).

In order to comply with the Environmental Protection Agency regulations in the US, the bleaching sequences were changed from using chlorine gas as the primary bleach agent to using chlorine dioxide, peroxide, or ozone (Brooks et al., 1994), (Blomback et al., 1998), (Fiskari et al., 1997), (Adams, 1995). The change in bleaching agent aimed to incorporate those filtrates into the black liquor cycle without corroding boiler tubes or increasing process discharge as wastewater. The process discharges are deemed elemental chlorine-free (ECF) if chlorine dioxide (ClO₂) is used and totally chlorine-free (TCF) if no chlorine compound is used in place of the chlorine-containing bleaching agents (Bajpai, 2012a). Sodium spiking is technically possible in both ECF and TCF processes, however, the potential for CO₂ capture is much greater in the TCF process; sodium spiking is only possible in ECF if an

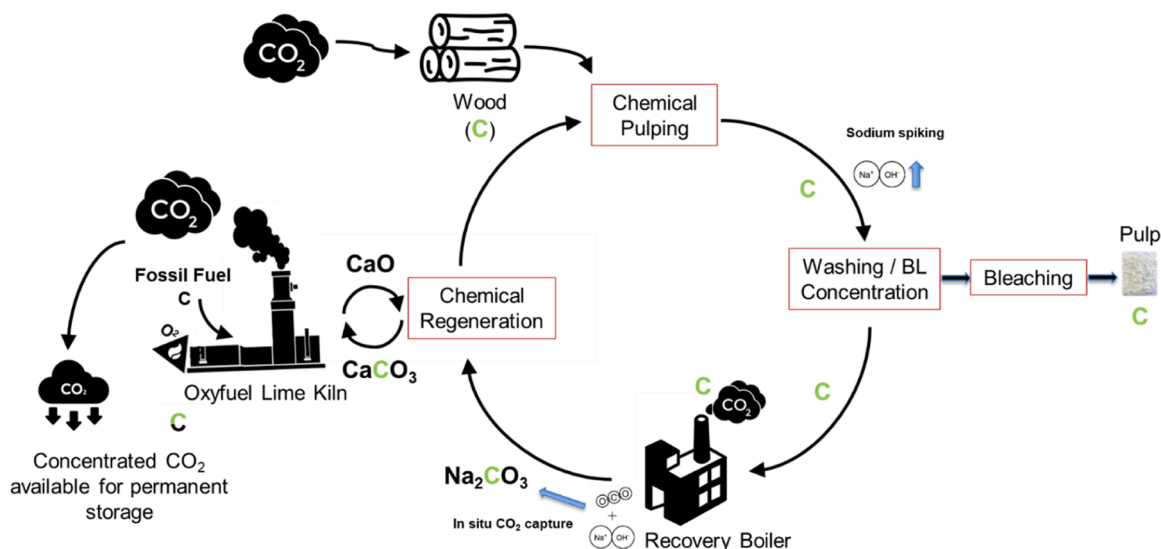


Fig. 1. Simplified process diagram for an integrated pulp mill with sodium spiking and oxyfuel combustion in lime kiln.

O₂ delignification stage is used prior to ClO₂ stages. Thus, TCF is the preferred pathway for sodium spiking. This study considers two scenarios: ECF and TCF. ECF bleaching sequences represent the primary industrial bleaching method utilized in paper mills globally (Bajpai, 2012b). In the US, ECF-based production accounts for 99% of the total bleached chemical pulp output (Fisher International, 2023). The ECF scenario is considered the reference without sodium spiking, wherein no bleached filtrates are returned to the chemical recovery cycle. The TCF scenario does incorporate sodium spiking.

In this study, we identify a valuable opportunity to increase the quantity of biogenic CO₂ generated in lime kilns at existing biomass pulping mills. Specifically, we propose a "sodium spiking" process wherein bleaching alkaline filtrates are recycled into the pulp-washing system. Sodium spiking increases the sodium (Na) content in the kraft pulping process, thereby increasing the quantity of CO₂ mineralized as Na₂CO₃ in the recovery boiler, as shown in Fig. 1. An increased generation of Na₂CO₃ causes an increase in CaCO₃ calcination and more high-purity CO₂ available for capture from the lime kiln.

Notably, the pulp and paper industry has already used the recycling of bleaching filtrates, or "sodium spiking," to increase economic viability and decrease waste effluents, but there has yet to be an investigation into the possibility of increasing the rate and decreasing the costs of carbon capture. A major advantage of integrating sodium spiking into biomass pulping is the reduced consumption of toxic and corrosive chlorine-bleaching compounds (Al-Wohoush et al., 2006). The bleached filtrate can be recycled into the brown stock washing system and combined with the black liquor to be burned in the recovery boiler, reducing the effluent charge as long as the filtrate contains no chlorine (Klarin and Westermarck, 1994).

For the first time, we demonstrate the potential of sodium spiking via TCF as a means to increase the rate and decrease the cost of CO₂ capture at biomass pulp mills. In addition, we identify bottlenecks in the biomass pulping process that limit the extent of sodium spiking and offer recommendations that will increase the probability of success when integrated into existing pulp mill operations.

2. Methodology

This work evaluated the effect of capturing CO₂ via sodium recycled from the alkaline bleaching stages to the recovery cycle of a hypothetical bleached pulp production process using loblolly pine as a feedstock. The performance of the chemical recovery cycle was assessed using a built model in the software WinGEMS v5.3 to provide thorough mass and energy balances in each unit operation involved in the bleached pulp production.

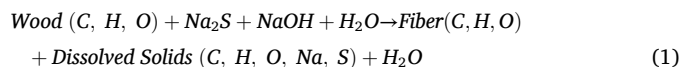
2.1. Southern bleached softwood kraft pulp mill

WinGems 5.3 is an iterative mass and energy balance software designed specifically for the Pulp and Paper industry. Therefore, the calculations are inherent in the system and based on some user assumptions. The WinGEMS model accounts for a baseline of 118 oven-dry metric tons (o.d. mt) of wood (roundwood and chips) per hour for a production of 43.8 oven-dry metric tons of southern bleached softwood Kraft pulp (SBSK). This production level is comparable with the daily average production rate (900 tons per day) for a typical bleached pulp mill in the United States (Fisher International, 2023). Table S1 shows the conditions used in the model.

Wood is made mainly of cellulose, hemicellulose, and lignin. From an elemental composition point of view, carbon (C), hydrogen (H), and oxygen (O) represent the conformation of wood, plus a small percentage of nitrogen and ash compounds that are negligible for modeling purposes (Andrews et al., 2015). Producing white pulp from wood requires combining steps that conform to a bleaching sequence that removes the residual lignin and the color structures in the pulp after the kraft pulping process. The Kappa value represents the amount of lignin that remains in

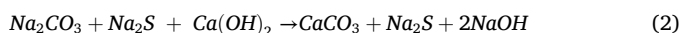
the fibers after the pulping process.

The wood is cooked in the digester under specific conditions (170°C and 160 psig) using white liquor to extract the fibers. White liquor mainly comprises sodium hydroxide (NaOH) and sodium sulfide (Na₂S). Eq. (1) summarizes the kraft pulping process. The liquor spent after cooking is called black liquor because of its dark appearance. It contains mainly dissolved lignin coupled with the inorganic compounds of the white liquor. In addition, the fiber is screened and separated from the black liquor to be washed. The wash water used in this stage goes back to the evaporator sets. Once cleaned, the pulp at medium consistency (12% solids) enters the bleaching stages.



Two evaporator sets remove water and concentrate the solids in the weak black liquor coming from the washing system. The first set works from 16% to 30% solids; the second ranges from 30% to 55%; and finally, a concentrator achieves 72% solids before being combusted in the RB. Inorganic ashes from the RB precipitator are mixed with the black liquor and then sent back to minimize the losses of inorganic species. The black liquor combustion in the RB fulfills three essential functions: to recuperate the inorganic compounds used in kraft pulping as smelt, the appropriate disposal of unwanted lignin, and third, the use of the heat value from the organic content of the black liquor to produce steam and, subsequently, electricity.

After the black liquor burning process, the remaining product is the molten smelt at the bottom of the boiler, which contains the inorganic compounds that enter with the black liquor. Smelt flows into the dissolving tank to be diluted. That solution is named green liquor because of its slightly green appearance. Green liquor mainly comprises sodium sulfide (Na₂S) and sodium carbonate (Na₂CO₃). The reactions inside the recovery boiler that form sodium sulfide are thoroughly described in the literature (Damasceno et al., 2020), (Wag et al., 1997). Conversely, the mechanisms driving the formation of sodium carbonate in the recovery boiler need to be better understood. Eq. (2) summarizes the chemistry involved in the recovery cycle after the smelt leaves the RB (note that the carbon in the Na₂CO₃ and CaCO₃ is entirely biogenic) to produce the white liquor again:



Eqs. (3) and (4) represent the calcium cycle needed to regenerate the sodium hydroxide in the white liquor:



The green liquor is clarified and mixed with the lime (CaO) from the lime kiln in re-causticizing units, where the sodium carbonate reacts with slaked lime (Ca(OH)₂), producing sodium hydroxide (NaOH) and lime mud (precipitated calcium carbonate (CaCO₃)) (Eq. (2)). The sodium sulfide does not take part in any reaction in this zone. Then, the sodium hydroxide and sodium sulfide combine to form the white liquor, which is clarified to separate the solid calcium carbonate. The model controls the sulfidity and effective alkali of the white liquor with the makeup of sodium sulfate (Na₂SO₄) to the black liquor before feeding the recovery boiler and sodium hydroxide before storing the white liquor to be used in the digester, restarting the cycle. Lime mud from the bottom layer of the white liquor clarifier is washed and filtered to ensure a moisture content lower than 25% before feeding the kiln. At temperatures exceeding 800°C, lime mud is converted into calcium oxide (CaO), liberating carbon dioxide (Eq. (3)). This reaction occurs inside the lime kiln, where fuel (natural gas) is combusted to provide the required heat energy. Then, the quick lime (CaO) is sent to the causticizers again to continue the calcium looping role in producing white liquor.

The base case for this study is a conventional ECF bleaching sequence, which does not incorporate filtrate recycling into the black liquor cycle and, therefore, does not achieve sodium spiking. ECF was selected as the reference case because it is the dominant bleaching method used globally, making it the most representative baseline for current industrial operations in the pulp and paper industry. By using ECF as the starting point, this study ensures that the analysis remains relevant to existing mill configurations and industry practices. Additionally, comparing sodium spiking within a conventional ECF system provides a realistic benchmark for evaluating the feasibility of transitioning to TCF. Since TCF bleaching is generally considered less economically viable, demonstrating the potential benefits of TCF with sodium spiking—such as improved carbon capture economics—could make the transition more attractive to paper mills from both an economic and sustainability perspective. Although ECF configurations can adapt oxygen (O) delignification stages before the chlorine dioxide, this study considered the scenario without O stages as a reference to establish a baseline where no sodium is recycled. Therefore, the alternative cases consider a TCF sequence where the percentage of sodium hydroxide (NaOH) used in the oxygen delignification stage ranges from one percent to nine percent (based on oven-dry pulp), and the filtrate from the alkaline steps is returned to the weak black liquor cycle.

The utilized elemental chlorine-free (ECF) bleaching sequence consists of three units of chlorine dioxide (D) and two sets of alkaline extraction (E), where one of them is reinforced with peroxide (P). The totally chlorine-free (TCF) bleaching sequence consists of two oxygen delignification (O) stages, an acid stage (A), and two extraction stages, one with ozone and another with peroxide. Then, finally, a peroxide stage (De Assis et al., 2017). Fig. S1 presents the process flow diagram for each bleaching sequence.

After each stage, a washer cleans the pulp for the next step in the bleaching area. The ECF sequence has a "jump-stage" configuration, meaning fresh water is only used in the last chlorine dioxide stage (D₂) and the E stage. The filtrate from the D₂ stage washer is fed counter-current back to the D₁ and then the D₀ stage washer. Fresh filtrate used on the E stage returns to the EoP wash stage. Both small filtrate circuits go to the wastewater collector for treatment and do not join the weak black liquor circuit (open configuration).

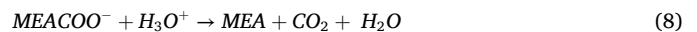
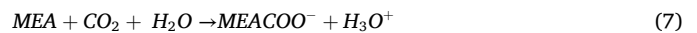
On the other hand, all the alkaline filtrates in the TCF sequence were sent back to the brown stock washing area, and thereby constituted sodium spiking in the recovery boiler. The main sodium hydroxide proportion is added in the oxygen delignification stage. Thus, the added freshwater is fed in the peroxide stage washer. The freshwater wash flow used in the model is controlled to keep 16% solids in the weak black liquor sent to the evaporators in all cases. The conditions and chemical

charges for the stages are listed in Tables S3 and S4.

The process variables (Table S1) correlate to a specific kappa number target in the unbleached pulp. The pulp yield, the composition of dissolved solids in black liquor, the chemical charge in the digester, and the efficiencies for significant areas such as the recovery boiler (reduction) and the causticizers (causticizing efficiency) are established for a given kappa number, and wood type (softwood). Generalized process flow diagrams are shown in Figs. 2 and 3; more detailed process parameters are in the supplementary information.

2.2. Liquid solvent-based carbon capture system

Lime kiln flue gas results from WinGEMS pulp mill simulation, under the different sodium spiking scenarios assessed, were used as input to perform a techno-economic analysis to establish the cost of CO₂ capture using an amine system. The amine system was simulated using Aspen Plus software (Adams et al., 2014), consisting of an absorber column where the CO₂ in the flue gas is in contact with the amine, and after being trapped in the monoethanolamine (MEA) solvent, it is sent to a stripping column where the CO₂ is released and the solvent is sent back to the absorber, as shown in Fig. S10. The resultant thermodynamic properties of the MEA-CO₂ contact are calculated using the ENRTL-RK model in Aspen. The molar composition of the lime kiln flue gas entering the absorber is CO₂ (19.2%), N₂ (47.8%), H₂O (31.8%), and O₂ (1.2%). The flue gas is cooled to 50°C before feeding the column. The features of the absorber and stripping columns using the rate-based model were determined following the methodology described by Madeddu et al. (2019). In both, the following kinetics reactions were implemented to control the interaction of CO₂ and MEA (Robano, 2024):



The rate of reaction (*r*) is calculated through the expression in Eq. (9), where *k* is the pre-exponential factor, *T* is the operating temperature in the columns, *E* is the activation energy, and *R* is the universal gas constant:

$$r = k * T * e^{\left(\frac{-E}{RT}\right)} \quad (9)$$

The kinetics parameters of the main reactions of MEA and CO₂

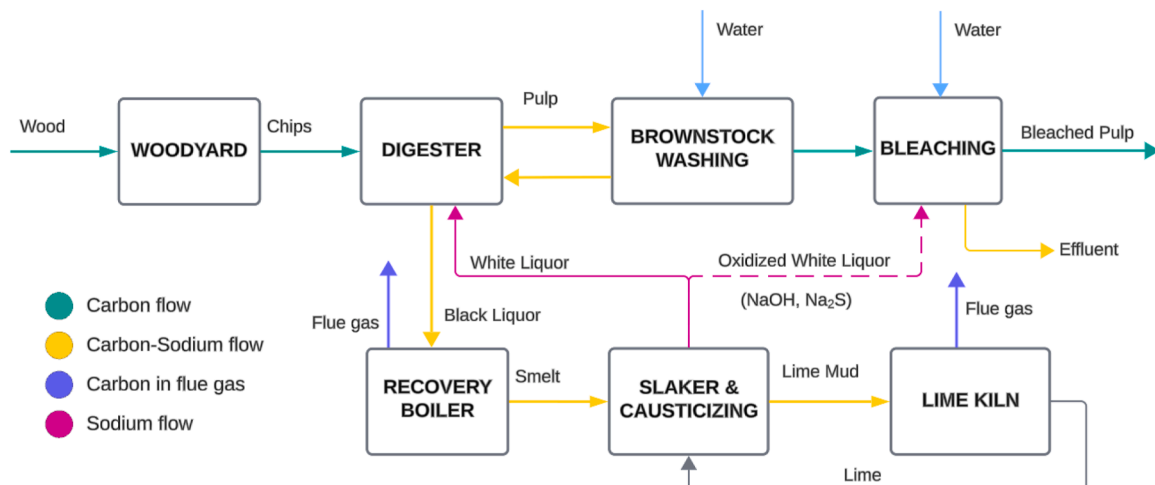


Fig. 2. Elemental chlorine-free (ECF) bleach process with chemical recovery.

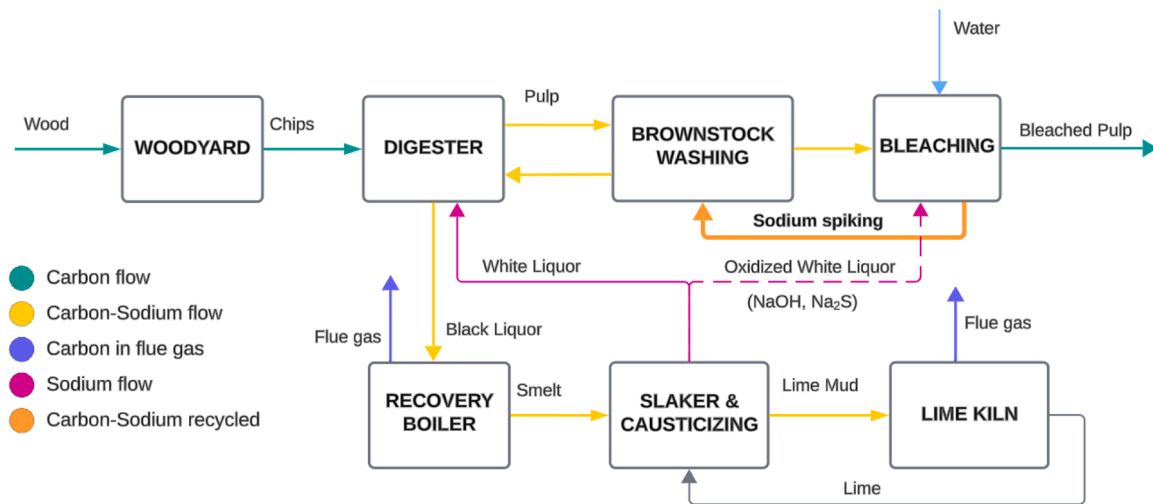


Fig. 3. Totally chlorine-free (TCF) bleach process with chemical recovery.

involved in the absorption and solvent regeneration are presented in Table 1.

A 90% capture rate was assumed for the MEA solvent-based captured system modeled according to previous studies (Onarheim et al., 2017), (Nwaoha and Tontiwachwuthikul, 2019), (Robano, 2024).

2.3. Oxy-fuel combustion and lime kiln retrofit

The techno-economic analysis also considered a retrofit to oxy-fuel combustion in the lime kiln to establish the cost of CO₂ capture. The lime kiln was modeled using Aspen Plus software under an oxy-fuel combustion process. A Rgibbs reactor was used to perform the calcination (Eq. (3)) of the lime mud coming out of the causticizers. The combustion of natural gas provided the heat to promote the transformation of CaCO₃ into CaO, releasing CO₂. An oxy-fuel combustion system replaces air with pure oxygen for combustion, eliminating the large amount of nitrogen that typically enters the kiln when using air. In traditional air-fired systems, nitrogen dilutes the combustion gases and absorbs heat, affecting overall thermal efficiency. In contrast, an oxy-fuel environment consists primarily of CO₂, H₂O, and residual O₂, which alters the heat transfer characteristics within the kiln. To maintain the desired kiln temperature and optimize heat distribution, a portion of the flue gas is recirculated back into the kiln. This recirculation increases the CO₂ concentration inside the kiln, creating a high-purity CO₂ environment that facilitates efficient separation of CO₂ from the exhaust stream without requiring an energy-intensive capture system. A 99% of the incoming CO₂ capture rate is assumed under oxy-fuel conditions (Talei et al., 2024).

3. Results and discussion

ECF and TCF bleaching scenarios were evaluated for the southern bleached softwood Kraft (SBSK) pulp production process. The following results and discussion are based on the SBSK pulp mill model. The ECF scenario serves as the reference case and does not send alkaline filtrate

Table 1
Kinetics parameters for the reactions of MEA and CO₂ used in Aspen modeling.

Reaction	k	E (cal/mol)	Refs.
5	1.33e+17	13,249	(Pinsent et al., 1956)
6	6.63e+16	25,656	(Aspen, 2008)
7	3.02e+14	9,855.8	(Hikita et al., 1977)
8 (Absorber)	5.52e+23	16,518	(Aspen, 2008)
8 (Stripper)	6.50e+27	22,782	(Aspen, 2008)

from the bleaching washers to the recovery boiler, thus excluding sodium spiking. In contrast, the TCF scenario includes sodium spiking by returning alkaline filtrate to the recovery boiler for in-situ CO₂ capture.

3.1. Carbon flow

Process simulation allows for the tracking of carbon and sodium flows inside the system boundaries of the typical US chemical pulp selected for analysis in this study (woodyard, digester, brown stock washing, bleaching, recovery boiler, and causticizing area, including the lime kiln). For example, [Fig. 4](#) shows the carbon flows for an SBSK pulp mill process with an ECF bleaching sequence to produce Softwood Kappa 30. Two inputs of carbon are taken into account: the carbon entering with the wood (~88.5% of total carbon) and the carbon via natural gas, used in the natural gas boiler (~9.5% of total carbon) and in the lime kiln to provide the energy to keep the high temperature required for the lime calcination (~2% of total carbon). Therefore, the majority of carbon is biogenic. The breakdown of carbon outputs from the ECF pulp mill includes 55.7% via carbon dioxide emissions in the flue gas of the recovery boiler, natural gas boiler, and lime kiln, 37.3% via fiber (final product) and wood rejects, and 7% as losses, as shown in [Fig. 4](#).

In the case of the TCF process, the carbon flow is shown in Fig. 5 for the scenario of a 3% sodium hydroxide charge in the oxygen delignification stage, a practical soda usage in industrial bleaching applications.

For both ECF and TCF processes, the carbon entering with the wood is the same. The carbon from the fuel used for the lime kiln is slightly higher in the TCF case due to the increase in sodium in the chemical recovery cycle, which demands a higher consumption of natural gas in the kiln. However, the total natural gas demands decreased significantly due to the lower steam required in the bleaching sequence, mainly due to the heat recovery during the recirculation of hot filtrate streams in the bleaching towards the brown stock washing system. The total steam demand in each process is summarized in Table S5. The biogenic carbon entering the process accounts for ~95% of the total carbon input in the TCF process. Importantly, in the TCF scenario, a portion of the carbon is returned from the bleaching area to the brown stock washing and finally integrated into the black liquor cycle, ending in the recovery boiler. In transitioning from ECF (Fig. 2) to TCF (Fig. 3), the load in the boiler is typically increased by 2 or 3% by recycling dissolved organics from wood into the recovery boiler (Bajpai, 2012f), (Bajpai, 2012d), (Bajpai, 2012e), (Bajpai, 2012c). The carbon leaving the TCF process in the flue gases accounts for 53.1% of the incoming carbon, 40.4% ends as fiber or wood rejects from the woodyard, and 6.5% is lost. The carbon losses in the TCF are lower relative to the ECF process. As can be seen, the

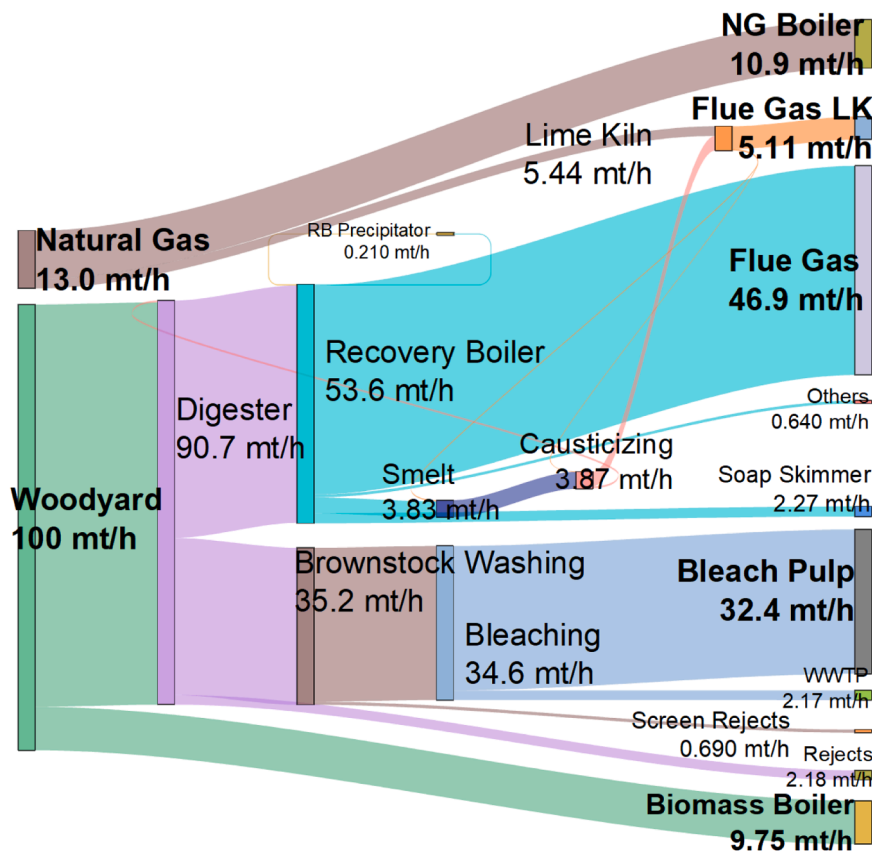


Fig. 4. Carbon flows for a southern bleached softwood kraft (SBSK) pulp mill process with an elemental chlorine-free (ECF) bleaching sequence without sodium spiking.

differences in carbon output between the bleaching sequences reside in the source of carbon emissions. Relative to the ECF process, if we compare the carbon output against the carbon input, the carbon emissions from the recovery boiler in the TCF scenario decrease. Fig. S4 compares the amount of carbon that leaves with the RB flue gas and the amount of carbon leaving the RB with the smelt for all the scenarios evaluated. As mass is not created or destroyed, the carbon is transformed from carbon dioxide in the recovery boiler to sodium carbonate and from sodium carbonate to calcium carbonate, which reacts in the lime kiln, releasing carbon dioxide at the highest concentration found in the kraft process.

3.2. Sodium flow

The balance of sulfur and sodium is a critical factor in the kraft process in terms of chemical recovery. Relative to the ECF process, the TCF process increases the amount of NaOH in the recovery cycle. Thus, additional sodium sulfide (Na_2S) must be compensated to conserve the correct balance between those compounds in the white liquor. When chemical adjustments are made to preserve a specific loading of Na_2S in the white liquor, the term used to refer to it is sulfidity, and when the objective is to conserve a defined amount of NaOH in the white liquor, the term used is causticity. Sodium makeup in the TCF process is achieved at two points: First, sodium sulfate is added to the concentrated black liquor before entering the recovery boiler, where it is reduced to promote Na_2S formation. Second, sodium hydroxide is added after the white liquor clarifier to control the sulfidity to around 30%.

Regarding the sodium balance, the causticizers, digester, and recovery boiler are the main areas for the ECF case:

1. The causticizing area produces the sodium hydroxide utilized in the digester.
2. The brown stock washers separate the dissolved solids (including sodium compounds) from the pulp as black liquor.
3. The sodium in the black liquor is sent to the recovery boiler, which is reduced into Na_2S and carbonated into Na_2CO_3 before being sent back to the causticizing area.

In this scenario, the sodium losses are mainly represented in the dissolved solids of the filtrate sent to the wastewater treatment from the bleaching area and the carryover in the bleached pulp. Fig. S2 shows the sodium flow for ECF Bleached Pulp from Softwood Kappa 30 (SBSK), wherein a vast majority of the Na is recycled.

In the case of the TCF process, the sodium flow is shown in Fig. S3 for the same scenario presented for the carbon balance (a 3% sodium hydroxide charge in the oxygen delignification stage). Here, the essential factor is the sodium hydroxide recycled from the alkaline bleaching stages to the weak black liquor. Fresh sodium hydroxide should not be used as the alkali source for bleaching because it would exceed the amount needed for sodium makeup in the chemical recovery system. This imbalance between the sodium and sulfur compounds would make managing the sulfidity in the white liquor more challenging.

Our model uses oxidized white liquor as an alkali charge in all the alkaline stages to preserve the chemical species balance. However, the sodium sulfide in the white liquor harms the brightness and viscosity of bleach pulp (Choudhury et al., 2011). Therefore, white liquor oxidation aims to convert the sulfide into sodium thiosulfate using air or pure oxygen. In addition, the capacity of the causticizing and lime-burning systems would need to be increased by around 5% if white liquor were used to fulfill the oxygenstage's whole alkali requirement (Dence and Reeve, 1996).

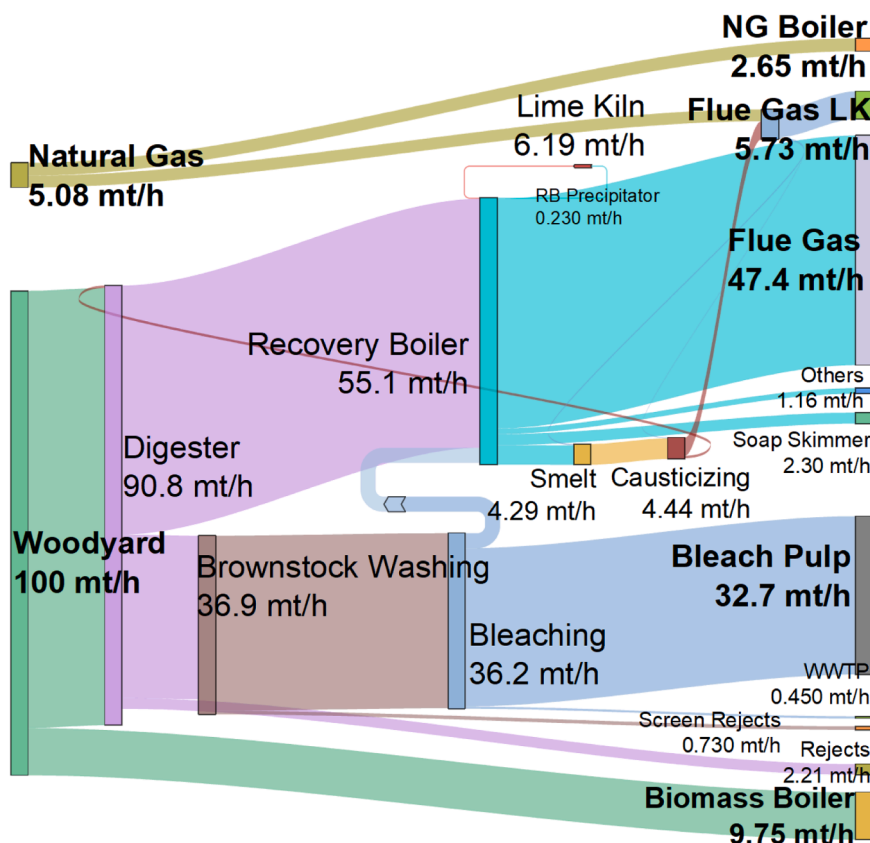


Fig. 5. Carbon flows for a southern bleached softwood kraft (SBSK) pulp mill process with a totally chlorine-free (TCF) bleaching sequence with sodium spiking.

3.3. Sodium spiking in a pulp mill

The process modeling is primarily focused on understanding the effect of increasing the quantity of sodium recycled from bleaching filtrates over the entire chemical recovery cycle. The amount of NaOH fed in the oxygen delignification stage is the only condition modified in the bleaching sequence, and it serves as a control variable to evaluate the amount of sodium increase in the bleaching filtrates. Typically, the oxygen delignification stage in TCF bleaching uses a range of 1.5 to 4.5% for the alkali (NaOH) charge. However, it was decided to increase the alkali charge beyond the normal range to maximize the extent of CO₂ mineralization in the recovery boiler and, subsequently, the extent of CO₂ capture from the lime kiln. As the NaOH concentration increases, less residence time is required for delignification, allowing for smaller-sized reactors and lower bleaching costs (Dence et al., 1996). The increase in alkali and the higher delignification rate also require fewer bleaching chemicals. The inorganic portion in the concentrated black liquor increased approximately one percent per each one percent of NaOH added in the oxygen delignification stages when filtrates are recycled.

The process model indicates that sodium spiking in the black liquor cycle—achieved by recycling alkaline filtrates from the bleach plant to the brown stock washing area—leads to an increase in carbon content within the recovery boiler smelt, as shown in Fig. S4. This elevated carbon content suggests a potential mechanism where sodium carbonate (Na₂CO₃) forms via in situ carbon dioxide (CO₂) capture. While some uncertainty remains regarding the exact reaction pathway, the overall process can be summarized by the following reaction:



Within the recovery boiler, organic compounds in the black liquor undergo thermal decomposition, producing CO₂, H₂O, CO, and H₂ gases, while inorganic sodium compounds are converted into Na₂CO₃ and Na₂S.

The oxidation of organics in black liquor provides a substantial and continuous source of CO₂ within the recovery boiler, further contributing to the potential for CO₂ mineralization reactions. Additionally, fume formation from inorganic species allows further interactions with CO₂. While some reactions can occur in the gas phase, previous studies (Hupa, 2007), (Frederick and Hupa, 1991) indicate that gas-phase Na₂CO₃ formation is minimal, as this reaction is significant only when nearly all the carbon in the char bed has been consumed, a point at which reduction reactions proceed very slowly. Therefore, the primary formation of Na₂CO₃ likely occurs within the char bed of the recovery boiler.

The literature extensively discusses sulfur reactions, particularly sulfate reduction within the char bed, as a key pathway for CO₂ and CO formation (Vakkilainen, 2006), (Bajpai, 2018), (Adams et al., 2019). These reactions are represented by the following equations:



The reaction in Eq. (12) requires 2.7 times more heat than in Eq. (11), making the reduction pathway that produces CO₂ more thermodynamically favorable and dominant in the recovery furnace (Adams et al., 2019). Since Na₂CO₃ formation requires an oxidized form of carbon, the abundance of CO₂ from both the oxidation of organics in black liquor and sulfate reduction makes it highly probable that CO₂ is first generated within the char bed and subsequently reacts with sodium species to form Na₂CO₃. This mechanism aligns with the observed increase in Na₂CO₃ content in the smelt when sodium spiking is implemented, reinforcing the proposed in situ mineralization process.

According to Eq. (10), for every two moles of sodium hydroxide added to the pulping process, one additional mole of carbon dioxide is captured and sent to the causticizers as sodium carbonate. As shown in Fig. S4, the total amount of carbon entering the RB increases with

elevated NaOH loading due to the dissolution of more organics during the bleaching sequence. In addition, the carbon contents of the smelt and flue gas increase and decrease, respectively, as more CO₂ is mineralized into smelt form.

In the ECF case, the smelt composition has the following distribution: 71% sodium carbonate and 17% sodium sulfide; the remaining balance comprises a small portion of sodium hydroxide, sodium sulfate, and inerts. In the TCF case, the proportion of sodium carbonate increases to 74%, with a slight decrease in the sodium sulfide percentage (16%) and the other compounds. Fig. 6a shows the additional amount (percentage) of smelt generated in each case compared to the ECF base case, where Na₂CO₃ and other sodium compounds are generated.

The generation of Na₂CO₃ in the smelt increases by four percent per each one percent of NaOH added in the oxygen delignification area. However, process model simulations suggest that at higher NaOH loadings, sodium carbonate formation levels off, while other sodium compounds, such as Na₂S, become more dominant. As illustrated in Fig. 6a, the relative abundance of Na₂S increases with higher NaOH additions, while Na₂CO₃ stabilizes (Fig. 6b).

Thus, recycling alkaline filtrate clearly enhances CO₂ mineralization in the recovery boiler, and now attention shifts toward impacts on upstream and downstream operations and the integration of carbon capture technologies. With such a dramatic increase in Na content in the kraft cycle, there is an opportunity for negative impacts on auxiliary operations, particularly the evaporation units. Fortunately, process model simulations show that elevated carbonate contents in the black liquor due to sodium spiking are not high enough to cause fouling in evaporators (Chai et al., 2001). In addition, the heating value of the black liquor is mildly decreased due to the increase in sodium content, but with no significant consequence since the lower heating value is offset by the increase in the total flow rate of black liquor to the RB. The estimated thermal efficiency for the RB in the ECF is 62.8%, being slightly reduced to 61.8% and 60.1%, for 3% and 9% TCF cases, respectively. The overall capacity of the evaporator system and the recovery boiler would need to be increased by ~3% to accommodate the additional sodium in the system (Dence and Reeve, 1996). Since the recovery boiler is the central and most expensive equipment in the Kraft process, it typically represents the main bottleneck in pulp and paper operations (Di Francesco et al., 2021). Upgrades to increase their capacity require high capital investments and possible unproductive downtime (Axelsson et al., 2006). Therefore, it may be preferable for new or recently integrated paper mill operations to leave an unused buffer capacity in the recovery boiler to allow for an increase in the rate of black liquor burning in potential future production expansions

(Bajpai, 2017). Interestingly, prior studies have shown that it is possible to utilize computational fluid dynamic (CFD) simulations to increase RB flow with little capital expenditure (Salmenoja, 2019). As a result, sodium spiking can arguably be a palatable or even attractive system to integrate in any mill as CFD optimization has shown in real world applications to provide more than the 3% gain needed for sodium spiking (Verrill et al., 1997), (Maakala et al., 2018), (Salmenoja, 2019). As a result, the mill production bottleneck is found in the evaporators or lime kilns, where there is generally very little if any extra room for increased capacity. If needed, incorporating an evaporator body into the evaporation system is easier and cheaper than modifying the RB capacity (Axelsson et al., 2006). In the case of lime kiln as a bottleneck, conceptual research has shown that retrofitting LK to oxy-fuel combustion allows an increase in lime production without hurting lime quality (Granados et al., 2015). Thus, retrofitting lime kilns to full or partial oxy-fuel combustion should accommodate the increased capacity required for sodium spiking.

As the sodium carbonate in the smelt produced in the RB increases in the TCF scenarios, calcium carbonate production (Fig. 7a) is expected to be higher through the causticizing reaction (Eq. (2)). The white liquor must be regenerated from the smelt, and the calcium cycle, combined with the sodium cycle, is used to complement the recovery of chemicals. Calcium oxide (CaO) is used to produce calcium hydroxide (Ca(OH)₂) through hydration. Once hydrated, the calcium interchanges positions with the sodium in the green liquor, as Eq. (2) indicates, to precipitate calcium carbonate (CaCO₃) and regenerate the sodium hydroxide, producing again the white liquor needed in the wood cooking process. As stated earlier, sodium sulfide is inert in this part of the process.

The CaCO₃ that is precipitated by the causticizing reaction (Eq. (2)) is removed from the liquid in a clarifier. This precipitate, referred to as lime mud, is then washed and filtered to increase its solid content. Lime mud is then put into a rotating kiln, where thermal energy is used to produce CaO and CO₂, as shown in Eq. (3). The production of calcium oxide and carbon dioxide has a linear relationship with the amount of calcium carbonate that initiates the reaction, taking into account the efficiency, mass, and heat transfer limitations; the final production for both calcium oxide and carbon dioxide in each scenario is shown in Fig. S6a and S6b.

As shown in Table 2, an average pulp mill has the potential to increase the amount of CO₂ captured from the lime kiln from 95 kt-CO₂/year to 125 kt-CO₂/year with the incorporation of sodium spiking.

Fig. 7b projects the annual amount of carbon dioxide produced in the lime kiln for a standard kraft pulp mill when a capture system with 90% capture rate is utilized. In the case of a retrofitted lime kiln with oxy-fuel

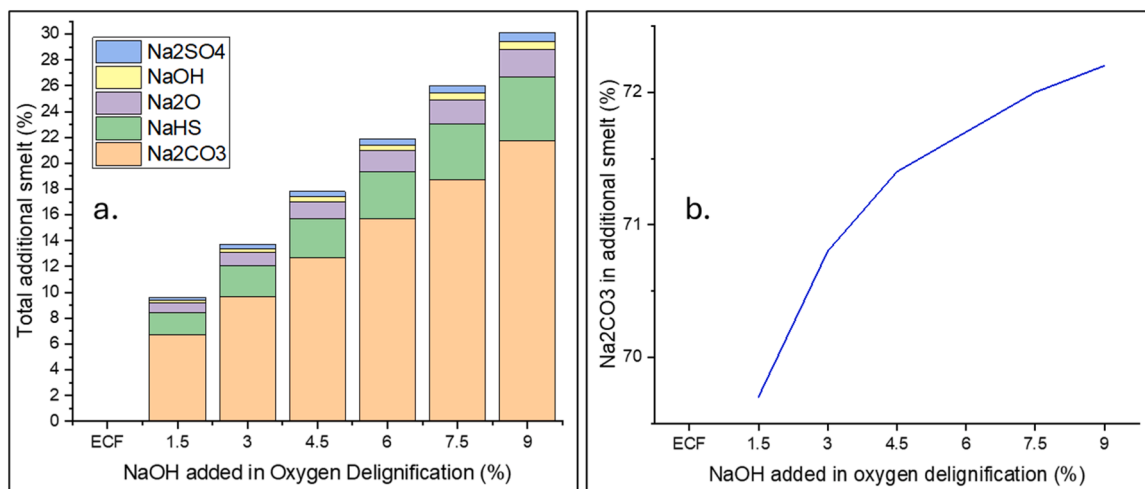


Fig. 6. a. Additional smelt production in the recovery boiler for each TCF scenario with sodium spiking as compared to the ECF scenario without sodium spiking. b. Extra sodium carbonate production in the recovery boiler for each TCF scenario with sodium spiking as compared to the ECF scenario without sodium spiking.

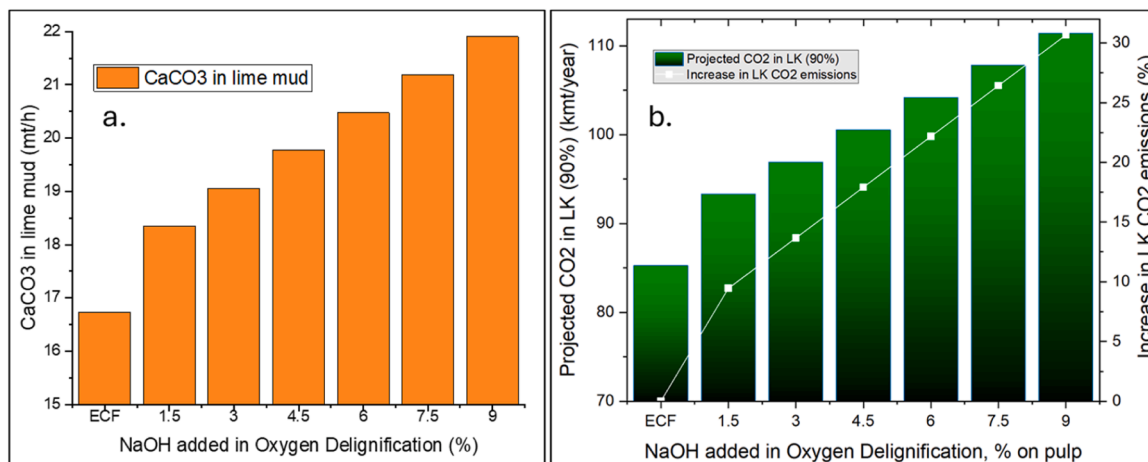


Fig. 7. a. Calcium carbonate production for each TCF scenario with sodium spiking. b. Annual projection of expected carbon dioxide capture in the Lime Kiln for each scenario.

Table 2

The increase in CO₂ emissions from the lime kiln with Na spiking.

Scenario	LK CO ₂ emissions (kT/year)	Percent increase in CO ₂ emissions
ECF	95.0	-
1.50%	104.5	9.5%
3%	108.0	13.7%
4.50%	112.1	17.9%
6%	116.3	22.2%
7.50%	120.5	26.4%
9%	124.7	30.7%

combustion, the potential CO₂ to capture is represented by the data in Table 2. The pulp and paper industry has immense potential in applying carbon capture and storage technologies in sources like boilers (natural gas, biomass, and recovery boiler) and the lime kiln. The company CO280 is actively implementing CCS in commercial pulp mills in the US (Aker Carbon Capture, 2024).

In the US, 46 pulp mills currently produce bleached pulp, and 27 out of 46 pulp mills are suitable for near-term implementation of sodium spiking (Fisher International, 2023). Among these, 16 mills are co-located with the suitable geology for CO₂ injection, as shown in Fig. S8 (Jennifer et al., 2023). Further, 23 of the 46 pulp mills incorporate oxygen delignification stages with ECF bleaching sequences, leading to sodium spiking if bleaching filtrates are being recirculated to the chemical recovery cycle. Incorporating sodium spiking and CCS in mills with existing oxygen delignification could be cost-attractive if the air separation units have extra capacity and lime kiln retrofitting to oxy-fuel combustion is feasible.

The incorporation of sodium spiking increases the load and throughput in the lime kiln operation, and thus lime kilns will need to have extra capacity or the ability to improve efficiency. If existing lime kilns are unable to accept the increase in load, then an alternative path could be through the use of full or partial oxy-fuel combustion. Retrofitting existing lime kilns to full or partial oxy-fuel combustion increases the capacity due to the increase in reaction kinetics (Granados et al., 2015). In addition, the elimination of nitrogen in the fuel mix produces a stream of carbon dioxide that is of much higher purity as compared to that produced using traditional combustion with natural gas. The high-purity CO₂ product does not require extensive downstream processing, thereby reducing costs. The energy demand for CO₂ capture has an inverse, non-linear relationship with CO₂ concentration; as CO₂ concentration increases, the energy demand for capture decreases exponentially (Wilcox, 2012). The generation of oxygen can create a parasitic load and extra cost for pulp mills, but costs may be minimal for mills that generate a surplus of onsite energy and/or have air separation

units already in operation with extra capacity; air separation is needed for O₂ delignification, which is typically used in TCF processes. Fig. 8 compares carbon dioxide concentrations in the flue gas between the current combustion and oxy-fuel combustion on a wet and dry basis. The combustion of natural gas (mainly methane, a standard fuel for the lime kiln) is defined in Eq. (13). An oxygen excess is needed to ensure complete combustion. Therefore, the outlet flue gas contains carbon dioxide, water, and oxygen.



Accordingly, the final concentration of CO₂ in the flue gas stream is high compared to regular air combustion, facilitating carbon dioxide capture after the water removal.

Retrofitting lime kilns from traditional air combustion to oxy-fuel combustion is a complex endeavor due to the change in internal environment. Oxy-fuel combustion elevates the CO₂ concentrations inside the kiln which in turn requires higher temperatures to achieve calcination (Wu et al., 2024). In addition, the kinetics of calcination (Eq. (3)) increase under oxy-fuel conditions, which will require higher throughput (Granados et al., 2015). Kinetic assessment using lime material from actual pulp mills and techno-economic modeling of lime kiln retrofitting are needed to better understand the feasibility of using oxy-fuel combustion to manage the increased throughput in the lime kiln due to sodium spiking.

3.4. Techno-economic analysis

To explore the potential economic impact of sodium spiking in the integration of carbon capture systems in the paper industry, a techno-economic analysis (TEA) has been performed. Six scenarios involving the Kraft process with the sodium spiking concept and lime kiln combustion type are assessed, as shown in Table 3. A liquid solvent-based absorption using monoethanolamine (MEA) was selected as carbon capture method, which is a well-known sorbent with high selectivity towards CO₂ and of high technology-readiness-level. In the case of oxy-fuel combustion in the lime kiln, no carbon capture method is needed since the concentration of the CO₂ with the flue gas is significantly high and only requires water separation.

The flue gas from the lime kiln in the ECF sequence described in this work is used as a baseline for the air combustion system scenarios, which are coupled with the MEA system to capture the CO₂. The ENRTL-RK thermodynamic property package in Aspen Plus was used to predict the interactions between the solvent and the CO₂ and the process model was rate-based (Aspen, 2008). The CO₂ product is dehydrated, purified, and compressed using a multistage of six compressor units and heat

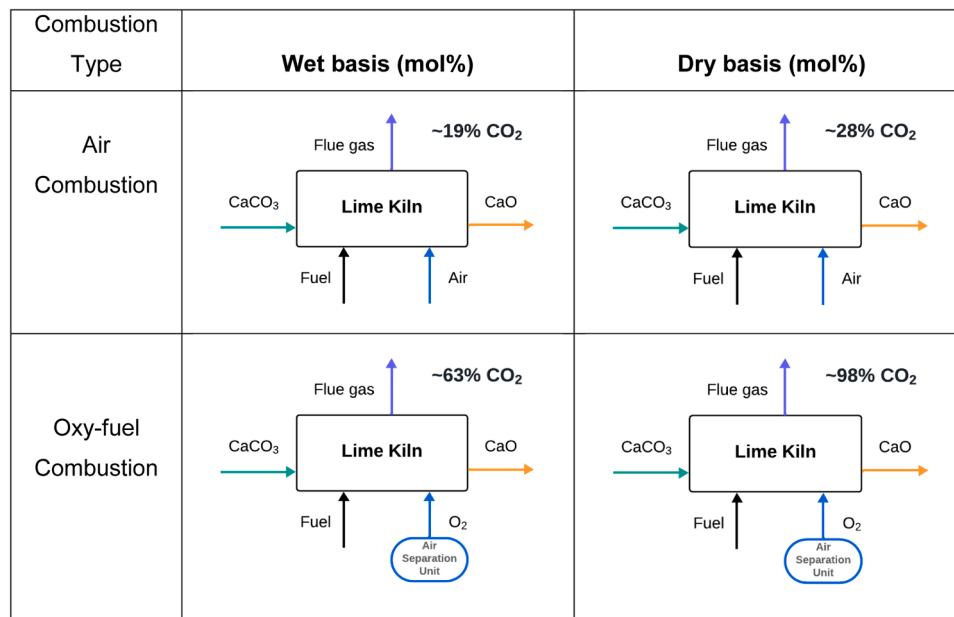


Fig. 8. Process flow diagrams comparing the carbon dioxide concentration from the lime kiln regarding the combustion type.

Table 3

Scenarios used to assess the techno-economic feasibility of CO₂ capture via sodium spiking.

Scenario	Bleaching sequence	Sodium spiking	Lime kiln
Baseline A	ECF	No	Air combustion
Baseline B	ECF	No	Oxy-fuel
Alternate 1A	TCF	Low*	Air combustion
Alternate 1B	TCF	Low*	Oxy-fuel
Alternate 2A	TCF	High*	Air combustion
Alternate 2B	TCF	High*	Oxy-fuel

Low: 3% NaOH charge in O₂ stages; High: 9% NaOH charge in O₂ stages

exchangers. The final conditions of the CO₂ are 153 bar and 44°C, where the CO₂ is in the supercritical state suitable for storage, pipeline transport, enhanced oil recovery (EOR) utilization, and/or geological sequestration (Mohitpour et al., 2012). The costs related to the geological storage of the CO₂ are not included in this evaluation.

In the case of the oxy-fuel combustion scenarios, due to the concentration of the resultant CO₂ in the flue gas, no amine system is used. Instead, an air separation unit (ASU) is incorporated based on the work of Keith et al. to provide the required oxygen to perform the calcination within 95% O₂ purity (Keith et al., 2018). In addition, a retrofit of the lime kiln is assumed to use pure oxygen as an oxidant to burn the natural gas and provide the heat required for the lime mud calcination (ECRA, 2012). The flue gas is dehydrated to produce a high-purity CO₂ stream. The same compression and purification units (CPU) are considered in these scenarios to reach the final CO₂ product specifications. Fig. 9 presents the schematic diagram flow for air and oxy-fuel lime kiln systems.

Table 4 presents the economic metrics used for this analysis, which are in accordance with the standard methodology presented by the National Renewable Energy Laboratory in their processes design reports (Dutta et al., 2011), and are consistent across all scenarios.

The initial equipment cost for the MEA and CPU system was generated using the Aspen Process Economic Analyzer (APEA) component for a flue gas rate of 48.3 metric tons per hour, and compared with previous studies (Jones, 2018). Then, to address the flow rate of flue gas inherent in each case scenario, the equipment cost was estimated using the investment scaling factor equation:

$$\frac{\text{Investment A}}{\text{Reference cost}} = \left(\frac{\text{Investment A capacity}}{\text{Reference capacity}} \right)^e \quad (14)$$

Where investment A is the desired capital cost to scale, reference cost is the cost of the device from the APEA, investment A capacity is the target capacity, reference capacity is the design capacity used in APEA, and *e* is the capacity factor. For the amine system and the CPU, a capacity factor between 0.7 and 1.0 was used, and for the oxy-fuel combustion a capacity factor of 0.9 was used (Onarheim et al., 2017), (Keith et al., 2018).

The cost of retrofitting the lime kiln was estimated according to the report presented by the European Research Cement Academy (ECRA, 2012). Future research about retrofitting kraft mills lime kilns to oxy-fuel combustion and the impact on product quality of CaO obtained from lime mud over an oxy-fuel lime kiln in the recovery chemical cycle is needed since most of the studies are based on cement clinker production.

Table 5 condenses the total capital investment for the CO₂ capture plant, the oxy-fuel combustion lime kiln retrofit, the CO₂ compression and purification system, the additional operational cost per year, and the amount of CO₂ captured. The additional operational cost comprises the cost associated with the chemicals for the new amine system, and additional chemicals make up compared to the corresponding baseline, additional electricity, natural gas demand, and labor costs. Table S6 presents a detailed breakdown of the capital and operational cost of integrating carbon capture and retrofitting the lime kiln to oxyfuel combustion under the different levels of sodium spiking in an SBSK mill.

Finally, Fig. 10 presents the cost of capturing each metric tonne of CO₂ for the six scenarios analyzed in this work and the detailed contribution of additional capital and operational costs. Without sodium spiking, the estimated CO₂ capture costs were \$132 and \$107/ mt CO₂ for air and oxy-fuel combustion, respectively. Similar costs have been reported for oxy-combustion systems in the cement industry (Nhuchhen et al., 2022), (Leeson et al., 2017). Previous studies have reported costs ranging from \$90 to \$150 when amine-based solvent systems are evaluated using the flow rates of a lime kiln in a kraft paper mill (Nwaoha and Tontiwachwuthikul, 2019) (Amod et al., 2022), (Onarheim et al., 2017). With the implementation of sodium spiking, the cost of CO₂ captured decreased by 27% and 31%, showing a cost of \$96 and \$74/mt CO₂ for air and oxy-fuel combustion, respectively. Oxy-fuel combustion

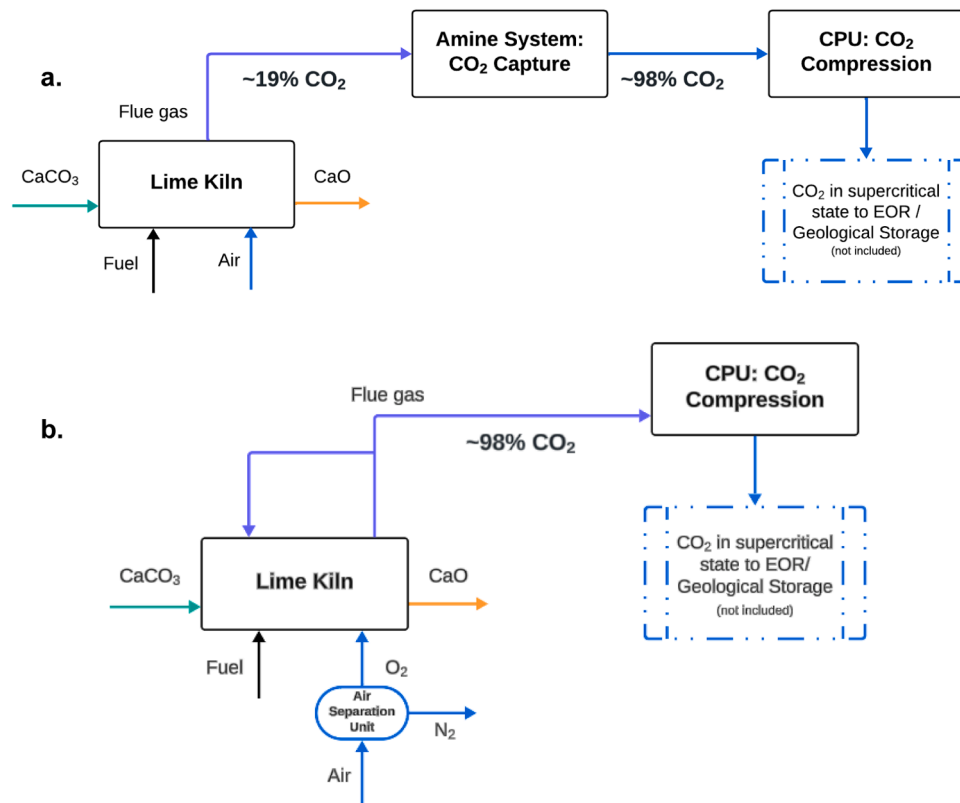


Fig. 9. Process flow diagrams for the CO₂ capture systems used in the techno-economic analyses. **a:** Amine system coupled to air combustion lime kiln, **b:** oxy-fuel combustion lime kiln.

Table 4

Baseline economic metrics for comparison of CO₂ capture costs.

Parameter	Value
Plant Utilization	96%
Cost year for analysis	2024
Equity financing	40%
Loan interest	8%
Loan terms, years	10
Depreciation period (years)	7
Construction period (years)	2.5
% spent in year 0	8%
% spent in year 1	60%
% spent in year 2	32%
Start-up time (years)	0.5
CO ₂ production at start up (% of normal)	50%
Variable costs (% of normal)	75%
Fixed costs (% of normal)	100%
Income Tax Rate	35%
Internal rate of return	10%

shows a good cost reduction potential compared to air combustion, but merged deployment with sodium spiking seems to be the most cost-effective pathway to integrate CCS. Additional economic benefits can be obtained if the CO₂ is stored and carbon credits are sold according to the voluntary markets, financial incentives or tax credits can be claimed from the carbon capture project, or revenue is obtained from the CO₂ utilization. Fig. S9 presents the cost of CO₂ capture for a case where CO₂ is used for EOR purposes.

Sodium spiking improves the throughput of CO₂ from the lime kiln available to capture, which decreases the capture cost per unit. Additionally, the steam demand of the entire process decreases due to the higher temperature of oxidized white liquor used to supply the sodium hydroxide to alkaline stages in the bleaching sequence. Hence, there is a lower demand of natural gas in the natural gas boiler with low sodium

spiking. However, we observed that in the scenarios where high sodium spiking was evaluated, the CO₂ capture cost increased. Chemical recovery cycle in kraft mills is a balance between sulfur and sodium moieties. When a higher dosage of sodium hydroxide is used, a higher makeup of chemicals is needed to balance the cycle. Table 5 includes the additional amount of NaOH, Na₂SO₄, and CaO required for each case scenario. Thus, the operational cost increases substantially for the high sodium spiking scenario as compared to low sodium spiking. Lime mud calcination under an oxy-fuel system in the pulp and paper industry could represent a cost-effective pathway to reducing carbon emissions. Lower capital and operational costs of ASU under oxy-fuel combustion compared to CCS in traditional air combustion scenarios compensate for the slightly higher compressors' capital cost and power consumption required. Similarly, the higher amount of CO₂ captured under oxy-fuel combustion mitigates the cost of additional make up chemicals required when sodium spiking is implemented. Due to carbon's biogenic nature in this industry, capturing the carbon emissions from lime mud calcination with an oxy-fuel system may be considered net carbon removal if the life cycle fossil emissions are lower than biogenic emissions capture and stored, which is an area that warrants future research.

4. Limitations

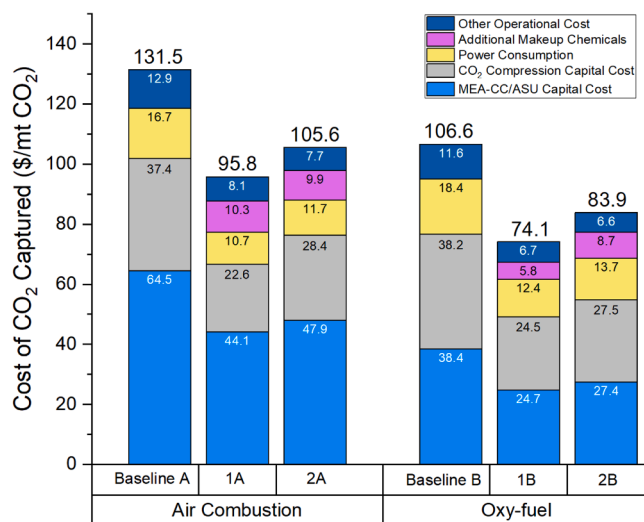
This study examines the impact of in situ biogenic carbon capture in the recovery boiler when sodium spiking is implemented. By recycling residual NaOH from the bleaching filtrates into the black liquor before combustion, more Na₂CO₃ is generated in the smelt. This, in turn, increases the biogenic CO₂ content in the lime kiln flue gas of a Kraft pulp mill. However, the extent of this effect can vary due to several factors, including washing system efficiency, the physical behavior of bed char in the recovery boiler's reduction zone (which is influenced by airflows), and other operational variables in the pulping process.

A techno-economic analysis was conducted to assess the cost

Table 5

Summary performance of scenarios evaluated.

Scenario	Total Capital Investment (TCI) (M\$)	Operating Costs (M \$/yr)	Biogenic-based CO ₂ captured (mt/yr)	Fossil-based CO ₂ captured (mt/yr)	Total CO ₂ captured (mt/yr)	Additional Chemicals makeup (mt/yr)
Baseline A	55.2	3.72	49,792	36,954	86,746	-
Baseline B	44.2	3.73	54,313	38,530	92,844	35,533
Alternate 1A	61.7	0.84	56,366	41,662	98,029	3,489,221
Alternate 1B	50.0	1.03	63,048	44,543	107,591	3,533,868
Alternate 2A	70.0	2.16	64,649	47,784	112,433	5,614,596
Alternate 2B	56.2	2.36	71,044	50,192	121,236	5,667,667
Amine (MEA) concentration (wt%)			30 %		(Dubois et al., 2017)	
Cost of MEA (\$ per tonne)			\$1,720		(Onarheim et al., 2017)	
Air separation unit power demand (kWh/ton O ₂)			238		(Keith et al., 2018)	
CO ₂ compression power demand (kWh/ton CO ₂)			132		(Keith et al., 2018)	
					(Sagues et al., 2020)	
Cost of electricity (\$ per kWh)			\$0.0751		(US Energy Information Administration, 2023)	
CO ₂ output condition (bar)			153		(Mohitpour et al., 2012)	
Total CO ₂ capture rate (%)			Air combustion	90	(Onarheim et al., 2017)	
			Oxy-fuel	99	(Talei et al., 2024)	

**Fig. 10.** Cost of CO₂ capture for baseline, low sodium spiking, and high sodium spiking scenarios.

implications of sodium spiking for CO₂ capture from the lime kiln. This analysis relies on capital cost estimates for different bleaching sequences based on prior studies (De Assis et al., 2017). However, actual costs can vary depending on site-specific mill conditions, and future work incorporating real-world data could improve cost estimations.

Regarding the carbon capture method, alternative solvents with higher capture efficiency than MEA (Nwaoha et al., 2018), (Xuechong Ding et al., 2023) could enhance the economic viability of integrating sodium spiking and CCS in existing air combustion kilns. Exploring these alternatives may further optimize the process for industrial application.

Additionally, the transition from ECF to TCF bleaching and the retrofit of the lime kiln combustion system introduce operational changes beyond CO₂ capture. While the reduction in natural gas usage for steam generation presents a notable cost-saving benefit, its impact on mill-wide power generation was not evaluated in this study. Future research using specific mill site data should refine the techno-economic feasibility of sodium spiking and oxy-fuel combustion retrofits in pulp and paper mills, providing a more accurate assessment of their overall benefits and trade-offs.

5. Conclusions

Lime kilns at pulp mills generate relatively high-purity biogenic CO₂ streams, making them ideal candidates for low-cost capture and sequestration. Sodium spiking is a novel technique that existing pulp mills can utilize to increase the amount of CO₂ captured in the lime kiln by recycling alkaline filtrates from the bleaching process into the pulp-washing system. The relatively high sodium content in the black liquor mineralizes CO₂ when burned in the recovery boiler. The mineralized CO₂ is then directed to the lime kiln, which is ultimately released in a much higher concentration and purity than the recovery boiler. Each one percent increase in sodium hydroxide used in upstream bleaching operations increases the downstream generation of CO₂ in the lime kiln by four percent. Thus, sodium spiking increases throughput in the lime kiln operation, which may require process intensification, including retrofitting to oxy-fuel if there is insufficient extra capacity in the lime kiln. Without sodium spiking, lime kilns' estimated CO₂ capture costs are \$132 and \$107/mt CO₂ for air and oxy-fuel combustion, respectively. With the implementation of sodium spiking, the cost of CO₂ captured decreases by 27% and 31%, showing costs of \$96 and \$74/mt CO₂ for air and oxy-fuel combustion, respectively. For the first time, a chemical pulping process with sodium spiking and oxy-fuel combustion retrofit in the lime kiln was shown to reduce CO₂ capture costs by 31%, which has the potential to be highly impactful given that there is significant incentive and motivation to capture biogenic CO₂ from industrial operations, including pulp and paper. Future work using specific mill site data should improve the accuracy of the techno-economic feasibility of incorporating sodium spiking and oxy-fuel combustion into existing pulp and paper mills for CO₂ capture. Furthermore, research into the techno-economic feasibility for net negative emissions from pulp mills with sodium spiking and retrofitted lime kilns should be explored.

CRedit authorship contribution statement

Edgar Carrejo: Writing – review & editing, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Hasan Jameel:** Writing – review & editing, Supervision, Methodology, Investigation, Formal analysis. **Sunkyu Park:** Writing – review & editing, Supervision, Investigation, Formal analysis, Conceptualization. **William Joe Sagues:** Writing – review & editing, Writing – original draft, Supervision, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

William Joe Sagues reports financial support was provided by US Department of Energy. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

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Data availability

Data will be made available on request.

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